Exp 2 · Half/full adder and subtractor; 7483 adder, 2's-complement subtractor, BCD adder

Manual pp. 5–9 · ICs 7486, 7408, 7432, 7404, 7483 · all expressions checked over every input

MANUAL EXP 2, AIM

Design and test a half adder/half subtractor and a full adder/full subtractor; study the 7483. Viva form: "How does a single 7483 subtract, and what does C_{out} tell you after a subtraction?"

GUESS FIRST

Subtraction A − B = A + (2's complement of B). How would you get B's 2's complement using XOR gates and the carry-in?

Designs

Circuit	Outputs
half adder	$S = A \oplus B$, $C = AB$
half subtractor (A − B)	$D = A \oplus B$, $B_o = A'B$
full adder	$S = A \oplus B \oplus C_{in}$, $C_{out} = AB + BC_{in} + AC_{in}$ (= $AB + C_{in}(A \oplus B)$)
full subtractor	$D = A \oplus B \oplus B_i$, $B_o = A'B + (A \oplus B)'B_i$

A B C_{in}	000	001	010	011	100	101	110	111
full adder S C_{out}	0 0	1 0	1 0	0 1	1 0	0 1	0 1	1 1
full subtractor D B_o (third input = B_i)	0 0	1 1	1 1	0 1	1 0	0 0	0 0	1 1

Wiring

Circuit	Wires (7486/7408/7432: gate 1 = pins 1, 2 → 3; gate 2 = 4, 5 → 6; gate 3 = 9, 10 → 8; gate 4 = 12, 13 → 11)
half adder	A → 7486-1 and 7408-1; B → 7486-2 and 7408-2. **S = 7486-3, C = 7408-3**
half subtractor	D = 7486 (A → 1, B → 2 → 3). A → 7404-1 (pin 2 = A'); 7404-2 → 7408-1, B → 7408-2; **B_o = 7408-3**
full adder: sum	7486 gate 1: A, B → 3; gate 2: pin 3 → 4, C_{in} → 5; **S = 7486-6**
full adder: carry (Fig 2.3)	7408: A, B → 3 (AB); B → 4, C_{in} → 5 → 6 (BC_{in}); A → 9, C_{in} → 10 → 8 (AC_{in}). 7432: 7408-3 → 1, 7408-6 → 2 → 3; 7432-3 → 4, 7408-8 → 5; **C_{out} = 7432-6**
full subtractor: borrow	7404: A → 1 → 2 (A'); 7486-3 (A ⊕ B) → 3 → 4. 7408: 7404-2 → 1, B → 2 → 3 (A'B); 7404-4 → 4, B_i → 5 → 6. 7432: 3 → 1, 6 → 2; **B_o = 7432-3**

The 7483: 4-bit adder (note the odd power pins)



VCC is pin 5 and GND is pin 12, not 16/8. Wiring power to 16 and 8 is the classic mistake with this chip.

Signal	A3 A2 A1 A0	B3 B2 B1 B0	C_{in}	S3 S2 S1 S0	C_{out}
pin	1, 3, 8, 10	16, 4, 7, 11	13	15, 2, 6, 9	14

4-bit adder / 2's-complement subtractor (Fig 2.6)

- Feed each B bit through an XOR whose other input is the control **M** (7486), and also send **M to C_{in}** (pin 13).
- **M = 0:** B passes unchanged, C_{in} = 0, so the 7483 computes **A + B**.
- **M = 1:** each B bit is inverted (1's complement) and C_{in} = 1 adds the 1, giving 2's complement. So the 7483 computes **A − B**.

Wire	
b0 → 7486-1, M → 2 → pin 3 → 7483-11	b1 → 7486-4, M → 5 → pin 6 → 7483-7
b2 → 7486-9, M → 10 → pin 8 → 7483-4	b3 → 7486-12, M → 13 → pin 11 → 7483-16
M → 7483-13 (C_{in})	A3..A0 → pins 1, 3, 8, 10

Try	Inputs	S3..S0	C_{out}	Meaning
7 + 5	M = 0, A = 0111, B = 0101	1100	0	12
7 − 5	M = 1, A = 0111, B = 0101	**0010**	**1**	+2. C_{out} = 1 means no borrow, result positive.
5 − 7	M = 1, A = 0101, B = 0111	**1110**	**0**	C_{out} = 0 means negative: 1110 is −2 in 2's complement

BCD adder (Fig 2.7)

- Add the two BCD digits in 7483 #1.
- If the sum is more than 9, or produced a carry, add 6 (0110) in 7483 #2.

The correction signal is:

$$K = C_4 + S_3S_2 + S_3S_1$$

Make S3S2 and S3S1 with two 7408 gates, then OR all three with two 7432 gates. Feed **0 K K 0** into B3 B2 B1 B0 of 7483 #2, with the first sum on its A inputs and C_{in} = 0. **K is the decimal carry.**

Example: 7 + 5. 7483 #1 gives 1100 (12 > 9), so K = 1. Then 1100 + 0110 = (1)0010. Output: carry **1**, digit **0010**, which reads as 12 ✓.

WRITE IN YOUR RECORD

Result: the truth tables of the half/full adder and half/full subtractor were verified. The 7483 added two 4-bit numbers; with the XOR/M arrangement it subtracted in 2's complement (7 − 5 = 0010, C_{out} = 1; 5 − 7 = 1110, C_{out} = 0). The BCD adder gave 7 + 5 = 1 0010.

Theory for the record: a half adder adds two bits (S = A ⊕ B, C = AB). A full adder adds three, including the carry-in (S = A ⊕ B ⊕ C_{in}, C_{out} = AB + BC_{in} + AC_{in}), and can be built from two half adders and an OR. Subtractors produce a difference and a borrow (half: D = A ⊕ B, B_o = A'B). The 7483 is a 4-bit binary adder with fast carry. Subtraction uses A − B = A + B' + 1, made by XOR-ing B with a control M and feeding M to C_{in}. For BCD addition, 6 is added when the binary sum exceeds 9 or produces a carry.

CHECK 1

(a) In the subtractor, why does C_{out} = 1 mean "no borrow"? (b) BCD 8 + 9: give 7483 #1's output, K, and the final output. (c) A student powers the 7483 on pins 16 and 8. What happens?

Answers

Opening question. XOR each B bit with M (M = 1 inverts it) and feed M into C_{in} (+1). Together that adds the 2's complement of B, so the adder subtracts. After a subtraction, C_{out} = 1 means A ≥ B (positive result); C_{out} = 0 means the result is negative, in 2's-complement form.

Check 1. (a) A + (2^4 − B) = 2^4 + (A − B). When A ≥ B this overflows 4 bits, so carry = 1. When A < B it doesn't. (b) 8 + 9 = 17: #1 gives C_4 = 1, S = 0001, so K = 1. 0001 + 0110 = **0111**, carry 1, i.e. **1 0111** = 17 ✓. (c) Pin 16 is B3 and pin 8 is A1, so the chip gets no power. If the real VCC/GND pins (5, 12) are left floating while inputs are driven, it misbehaves and can be damaged. The outputs are garbage.